EXP – 05

GO – BACK – N Program

Server

```
import java.net.*;

public class GBNServer {
   public static void main(String[] args) {
      try {
         DatagramSocket serverSocket = new DatagramSocket(7000);
         System.out.println("GBN Server started...");

         byte[] buffer = new byte[1024];
         int expectedSeq = 0;

         while (true) {
            DatagramPacket receivePacket =
                  new DatagramPacket(buffer, buffer.length);
            serverSocket.receive(receivePacket);

            String msg = new String(
                  receivePacket.getData(), 0, receivePacket.getLength());

            if (msg.equals("END"))
               break;

            int seqNo = Integer.parseInt(msg);
            System.out.println("Received Frame: " + seqNo);

            int ack;
            if (seqNo == expectedSeq) {
               System.out.println("Frame Accepted");
```

```
                ack = expectedSeq;
                expectedSeq++;
            } else {
                System.out.println("Frame Discarded");
                ack = expectedSeq - 1;
            }

            // Send ACK
            String ackMsg = String.valueOf(ack);
            DatagramPacket ackPacket =
                    new DatagramPacket(ackMsg.getBytes(), ackMsg.length(),
                            receivePacket.getAddress(), receivePacket.getPort());
            serverSocket.send(ackPacket);
        }

        serverSocket.close();

    } catch (Exception e) {
        e.printStackTrace();
    }
  }
}
```

Client

```
import java.net.*;
import java.util.Random;

public class GBNClient {
  public static void main(String[] args) {
    try {
```

```
DatagramSocket clientSocket = new DatagramSocket();
InetAddress serverAddress = InetAddress.getLocalHost();

int windowSize = 4;
int totalFrames = 10;
int base = 0;
int nextSeq = 0;

byte[] buffer = new byte[1024];
Random random = new Random();

while (base < totalFrames) {

    // Send frames within window
    while (nextSeq < base + windowSize && nextSeq < totalFrames) {
        System.out.println("Sending Frame: " + nextSeq);

        DatagramPacket sendPacket =
                new DatagramPacket(String.valueOf(nextSeq).getBytes(),
                        String.valueOf(nextSeq).length(),
                        serverAddress, 7000);
        clientSocket.send(sendPacket);
        nextSeq++;
    }

    // Receive ACK
    DatagramPacket ackPacket =
            new DatagramPacket(buffer, buffer.length);
    clientSocket.receive(ackPacket);

    int ack = Integer.parseInt(
```

```
                new String(ackPacket.getData(), 0, ackPacket.getLength()));

            System.out.println("ACK Received: " + ack);

            if (ack >= base) {
                base = ack + 1;
            } else {
                System.out.println("Go-Back-N: Retransmitting from frame " + base);
                nextSeq = base;
            }
        }

        // End signal
        DatagramPacket endPacket =
                new DatagramPacket("END".getBytes(), 3,
                        serverAddress, 7000);
        clientSocket.send(endPacket);

        clientSocket.close();
        System.out.println("Transmission Completed");

    } catch (Exception e) {
        e.printStackTrace();
    }
  }
}
```

```
Command Prompt
ACK = 1 (Accepted)  Count = 10
All packets received successfully.

D:\Share\A27>javac GBNServer.java

D:\Share\A27>java GBNServer
GBN Server started...
Received Frame: 0
Frame Accepted
Received Frame: 1
Frame Accepted
Received Frame: 2
Frame Accepted
Received Frame: 3
Frame Accepted
Received Frame: 4
Frame Accepted
Received Frame: 5
Frame Accepted
Received Frame: 6
Frame Accepted
Received Frame: 7
Frame Accepted
Received Frame: 8
Frame Accepted
Received Frame: 9
Frame Accepted

D:\Share\A27>
```

```
Command Prompt
ACK = 1  Success Count = 10
Transmission completed.

D:\Share\A27>javac GBNClient.java

D:\Share\A27>java GBNClient
Sending Frame: 0
Sending Frame: 1
Sending Frame: 2
Sending Frame: 3
ACK Received: 0
Sending Frame: 4
ACK Received: 1
Sending Frame: 5
ACK Received: 2
Sending Frame: 6
ACK Received: 3
Sending Frame: 7
ACK Received: 4
Sending Frame: 8
ACK Received: 5
Sending Frame: 9
ACK Received: 6
ACK Received: 7
ACK Received: 8
ACK Received: 9
Transmission Completed

D:\Share\A27>
```